Sensor Importance for Observability Degree via Shapley Values*

**Independent research unrelated to Intuit.*

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Presentation Overview

1. Motivation & key contribution(s)
2. Mathematical preliminaries
3. Observability
4. Degree of observability
5. Shapley values
6. Shapley values for sensor importance
7. Results & conclusions

Motivation

- Modern systems (e.g., ADAS, power plants, aircraft, etc.) are heavily instrumented with sensors
  - Important to know *exactly how much* each sensor contribute
  - Cost/complexity vs. performance tradeoffs
  - Observability is often the criteria used to assess sensor importance (for estimator design)
- Degree of observability moves beyond observable/non-observable checks
- Current gaps/challenges
  - Sensor selection problems have combinatorial complexity
  - Sensor importance methods (Ruggaber *et al.*, 2021) require the implementation of a Kalman filter

Key contributions

- Inspired by Machine learning
  - Shapley values widely used for feature downselection and importance ranking
  - Key for explainable AI/ML
- Treat observability as a co-operative game
  - *Sensors* = players
  - *Payout* = degree of observability
  - *Shapley value* = fair, axiomatic sensor importance
- Contributions
  - First effort to leverage Shapley values to learn *fair* contribution of each sensor towards observability
  - Shapley value axioms provide *unprecedented* interpretability to sensor selection
  - Does not necessarily need the implementation of a filter

State space system

(Deterministic) discrete time linear time invariant (LTI) system:

$$\begin{aligned} \mathbf{x}_{k+1} &= A\mathbf{x}_k \\ \mathbf{y}_k &= C\mathbf{x}_k \end{aligned}$$

where

- $\mathbf{x} \in \mathbb{R}^n$ denotes the state vector
- $\mathbf{y} \in \mathbb{R}^p$ denotes the measurement vector
- $A \in \mathbb{R}^{n \times n}$ denotes the state transition matrix
- $C \in \mathbb{R}^{p \times n}$ denote the measurement matrix
- The time index is denoted by k and $k = 0, 1, \ldots, K$

Observability definition

Definition

A discrete time LTI system is said to be observable if the initial state $\mathbf{x}_0 \in \mathbb{R}^n$ can be uniquely recovered from the sequence of measurements (or sensors)[1] $\{\mathbf{y}_k\}_{k=0}^{K}$, where $K \geq n - 1$.

[1] Measurements and sensors are used interchangeably in this presentation.

Observability matrix

- Measurement equation at time index k

$$\mathbf{y}_k = CA^k \mathbf{x}_0$$

- For a finite horizon of $K \geq n - 1$ measurements

$$\begin{aligned} \begin{bmatrix} \mathbf{y}_0 \\ \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_K \end{bmatrix} &= \begin{bmatrix} C \\ CA \\ \vdots \\ CA^K \end{bmatrix} \mathbf{x}_0 \\ \mathcal{Y}_K &= \mathcal{O}_K \mathbf{x}_0 \end{aligned}$$

  - $\mathbf{x}_0$ is the unknown (initial) state to be estimated
  - Least squares problem

- $\mathbf{x}_0$ has a unique solution, if and only if the matrix $\mathcal{O}_K$ (*observability matrix*) has full column rank.

Observability Gramian

- An alternate approach to deducing observability is via the *Observability Gramian*.
- Energy of the output (measurement) over a horizon K is given by

$$
\begin{aligned}
\mathcal{E}_y = \sum_{k=0}^{K} ||\mathbf{y}_k||^2 &= \sum_{k=0}^{K} \mathbf{x}_0^T \left(A^T\right)^k C^T C A^k \mathbf{x}_0 \\
&= \mathbf{x}_0^T \left(\sum_{k=0}^{K} \left(A^T\right)^k C^T C A^k \right) \mathbf{x}_0 \\
&= \mathbf{x}_0^T \mathcal{W}_{\mathcal{O}} \mathbf{x}_0
\end{aligned}
$$

- Thus,

$$
\mathcal{W}_{\mathcal{O}} = \sum_{k=0}^{K} \left(A^T\right)^k C^T C A^k
$$

  - where $\mathcal{W}_{\mathcal{O}} \in \mathbb{R}^{n \times n}$ is known as the *observability Gramian* matrix.

Additive property

- Measurement matrix C in in terms of individual measurements (or sensors)

$$C = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_p \end{bmatrix}$$

  - where $C_i \in \mathbb{R}^{1\times n}$ represents the i^{th} row of C and $i = \{1, \ldots, p\}$.
- Additive property of observability Gramian

$$\mathcal{W}_{\mathcal{O}} = \sum_{k=0}^{K} \left(A^T\right)^k C^T C A^k = \sum_{i=1}^{p} \mathcal{W}_{\mathcal{O}}{}^i$$

  - where $\mathcal{W}_{\mathcal{O}}{}^i$ is the observability Gramian of the LTI sub-system (A, C_i)

To summarize...

- Observability $\implies$ $\mathcal{W}_{\mathcal{O}}$ must be positive definite.
  - Output energy has to be positive to recover the initial state $\mathbf{x}_0$.
- Observability Gramian $\leftrightarrow$ Observability matrix

$$\mathcal{W}_{\mathcal{O}} = \mathcal{O}_K^T \mathcal{O}_K$$

- Yes/No conditions:
  - Observability Gramian $\mathcal{W}_{\mathcal{O}}$: row and column rank n
  - Observability matrix $\mathcal{O}_K$: full column rank n

Degree of observability — Why?

- Conditions on observability matrix or Gramian are binary
  - Fully observable or unobservable
- Why is granularity needed?
  - Partial observability - Only some states are observable
  - How observable is a fully observable system?
    - Degree of observability quantifies this
    - Higher degree $\implies$ better convergence properties
- Applications in...
  - Sensor selection
  - Sensor placement
  - Virtual sensing (cost out)
  - Filter design (tuning)

Intuition for observability degree

- Least squares problem

$$\begin{aligned}\mathcal{Y}_K &= \mathcal{O}_K \mathbf{x}_0 \\ \mathcal{W}_{\mathcal{O}} &= \mathcal{O}_K^T \mathcal{O}_K\end{aligned}$$

- For initial state $\mathbf{x}_0$ to have a unique solution
  - The observability Gramian $\mathcal{W}_{\mathcal{O}}$ should be well-conditioned and invertible.
- Farther the matrix $\mathcal{W}_{\mathcal{O}}$ is from being singular, the better conditioned the problem is

Three categories of observability degree measures

- Modular – Linear function of observability Gramian.
  - E.g. trace of the observability Gramian – $Tr\{\mathcal{W}_{\mathcal{O}}\}$
- Sub-modular
  - E.g. log determinant of the observability Gramian – $\log|\mathcal{W}_{\mathcal{O}} + \lambda I|$
- Non-submodular
  - E.g. minimum eigenvalue of observability Gramian – $\lambda_{\min}\{\mathcal{W}_{\mathcal{O}}\}$

Shapley values – Mathematical preliminaries & intuition

- Concept developed for *fair* allocation of payouts in multi-player games
    - p players are playing a game
    - $v(\cdot)$ is the overall payout of the game
- Shapley values are a way to allocate payout to individual players
    - Expected marginal contribution made by players to the overall payout
- Coalition: Is a set of players who are playing a game
- Value or payout: Is a function that maps each coalition (or set) N of players to a (scalar) payout

$$v(\cdot) : 2^p \to \mathbb{R}$$

Shapley values definition

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} w(S) \left[v\left(S \cup \{i\}\right) - v\left(S\right) \right]$$

- S denotes a subset of the full set N that does not contain player i
- $N = \{1, 2, \ldots, p\}$ is the full set of players
- $w(S)$ is the weight attached to the set or coalition S
- $\phi_i(v)$ is the Shapley value of player i w.r.t value/payout function $v(\cdot)$
- The value function $v(\cdot)$ needs to be well defined for all possible coalitions $S \subseteq N$

Shapley value axioms

1. Efficiency: Shapley values of the individual players should add up to the overall payout of the game

$$\sum_{i \in N} \phi_i(v) = v(N)$$

where $N = \{1, 2, \ldots, p\}$ is the set of all players.

2. Symmetry: Players who contribute equally, get equal value

$$v\left(S \cup \{j\}\right) = v\left(S \cup \{k\}\right)$$

3. Dummy: A player who adds no value to any coalition gets zero

$$v\left(S \cup \{j\}\right) = v\left(S\right)$$

4. Additivity: If a game is addititive of multiple games, then the Shapley values of individual players are also additive

$$\phi_j(v_1 + v_2) = \phi_j(v_1) + \phi_j(v_2)$$
$$\phi_j(v^*) = \phi_j(v_1) + \phi_j(v_2)$$

Sensor importance from Shapley values

- Shapley value of a ***player*** quantifies fair contribution towards ***payout*** of ***game***
  - Player: Treat each sensor as a player, a group of sensors as a coalition
  - Payout: Treat observability degree measures as payout or the value function
  - Game: Coalition of sensors working together to improve overall observability degree
- Shapley value axioms automatically make the sensor importance measure interpretable
  - Applicable on *any well defined* observability degree measure
  - Combinatorial complexity – but excellent approximation techniques available
- Shapley values reward complementary sensors and penalize redundant sensors
  - Depends on value function chosen
  - They capture no additional information for modular functions

Simulation scenario I

Collaborative sensors

2-state, 2-measurement system

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad ; \quad C = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Sensor	Value Function	Standalone Value	Shapley Value
C_1	Trace	20.0	20.0
C_2	Trace	20.0	20.0
C_1	Min Eigenvalue	0	10.0
C_2	Min Eigenvalue	0	10.0

- Not observable if C_1 and C_2 used individually
- Shapley value observations
  - Symmetry axiom – both sensors are equally important
  - Efficiency axiom – Shapley values sum upto observability degree
  - Min. eigenvalue – Shapley values reward collaboration equally

Simulation scenario II

Complementary & redundant sensors

3-state, 4-measurement system

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad ; \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sensor	Value Function	Standalone Value	Shapley Value
C_1	Trace	3187.0	3187.0
C_2	Trace	295.0	295.0
C_3	Trace	5312.0	5312.0
C_4	Trace	10.0	10.0
C_1	Min Eigenvalue	1.3920	1.5129
C_2	Min Eigenvalue	0.0000	0.2306
C_3	Min Eigenvalue	0.6405	0.7089
C_4	Min Eigenvalue	0.0000	0.0243

- C_1 most important sensor and C_2, C_4 are redundant sensors
- Min. eigenvalue is able to correctly identify C_1 as most important followed by C_3
- C_2 and C_4 have low Shapley values as they contribute in coalitions

Conclusions

- Summary
  - Introduced a Shapley value–based notion of sensor importance for system observability
  - Provides fair, order-independent attribution that explicitly accounts for sensor interactions
  - Applicable to any observability metric (trace, determinant, minimum eigenvalue, etc.)
- Application
  - Turns observability from a binary/system-level concept into a diagnostic, sensor-level tool
  - Supports informed decisions in sensor prioritization, removal and redesign

The End

Questions? Comments?